Lecture notes: The Heat Equation

These notes overlap the slides, but also add PDF-only material.

Tick 'This lesson contains equations / formulas' when uploading.

1. Fourier's law

Heat flux is proportional to the temperature gradient.

$$q = -k \cdot T$$

The minus sign means heat flows from hot to cold.

2. The heat equation

Combining Fourier's law with conservation of energy gives:

$$\cdot u/\cdot t = \cdot\ \cdot^2 u$$

u is temperature, t is time, and $\cdot$ is thermal diffusivity.

This is the educator formula that must be copied exactly.

3. Superposition / series

On a finite interval with $u(0,t)=0$ and $u(L,t)=0$:

$$u(x,t) = \cdot\ a_n \sin(n\cdot x/L) \exp(-\cdot\ (n\cdot/L)^2\ t)$$

The spatial problem is $X'' + \cdot X = 0$.

4. PDF-only extra (should remain after dedupe)

Maximum principle: a solution of the heat equation attains its maximum on the parabolic boundary, not in the open cylinder.

Stability estimate:

$$||u(\cdot,t)||\_\cdot\ \cdot\ ||u(\cdot,0)||\_\cdot$$

Explicit Euler CFL restriction (PDF-only):

$$\cdot t\ \cdot\ (\cdot x)^2 / (2\cdot)$$

5. Compact notation

Energy identity: $d/dt\ \cdot\ u^2\, dx = -2\cdot\ \cdot\ |\cdot u|^2\, dx$